



Self-RAG in Enterprise AML

Authors Yanxuan Dong, Peiqi Liu

Affiliations Carnegie Mellon University

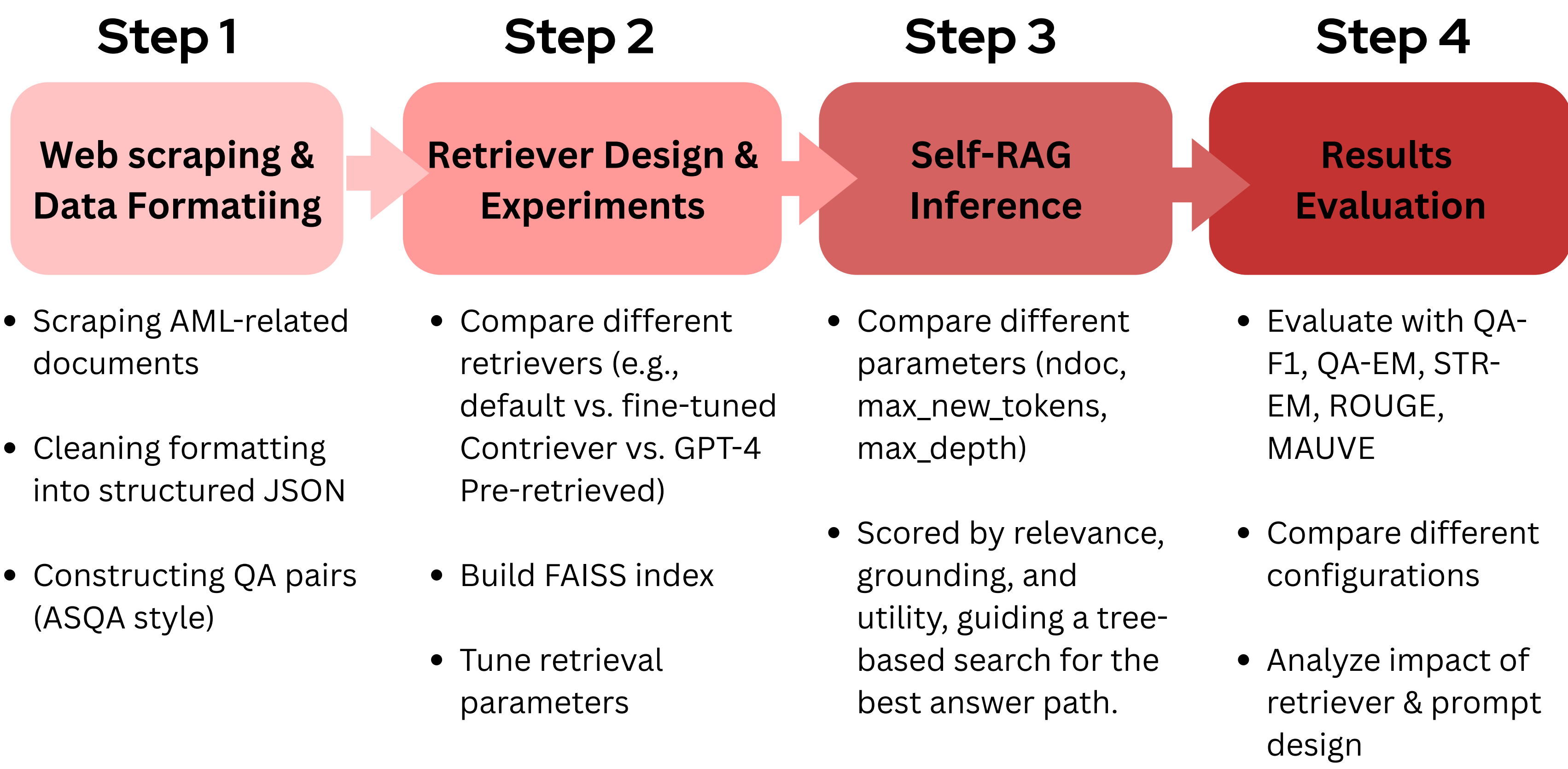
Introduction

Anti-Money Laundering (AML) is critical for financial enterprises to prevent fraud. However, accessing AML knowledge within large organizations is often time-consuming, requiring employees to manually search through scattered documents.

Our project aims to streamline this process by applying Self-Reflective Retrieval-Augmented Generation (Self-RAG) to enterprise-level QA. This allows employees to ask AML-related questions in natural language and receive accurate, grounded answers—without needing to locate the information themselves. It improves knowledge accessibility, reduces compliance risk, and enhances productivity in AML operations.

Methodology

Figure 1. Self-RAG in AML Workflow



Results

rougeLsum: GPT-4 pre-retrieval scores (42.52) significantly higher than the fine-tuned retriever (~37.5) and default retriever (35.3), indicating that the documents retrieved by GPT-4 produce better summaries.

QA-F1 and QA-EM: GPT-4 also performs better on these question-answering metrics, indicating that the documents it retrieves more directly contain the answer to the question.

mauve: Interestingly, the fine-tuned retriever is close to the default retriever (9.2) in terms of text naturalness (~9.5), while GPT-4 is lower (8.32).

	Fine-tuned Retriever + Simple Prompt--ndocs 4 --max_new_tokens 400 --threshold 0.2--max_depth 7"	Fine-tuned Retriever + Detailed Prompt--ndocs 5 --max_new_tokens 300 --threshold 0.2--max_depth 7"	GPT 4o pre-retrieval--ndocs 5 --max_new_tokens 300 --threshold 0.2--max_depth 7"	Baselinedefault retriever
length	26.4	23.49	29.86	32.99
str_em	1.77	1.65	9.1	1.35
str_hit	0.9	0.54	2.69	0.54
rougeLsum	37.56	37.45	42.52	35.3
QA-EM	0.24	0.24	1.2	0.6
QA-F1	10.8	11	14.36	9.22
QA-Hit	0	0	0.18	0.18
mauve	9.64	9.32	8.32	9.2

Figure 2. Experimental Results

Potential Improvements:

- Add more few-shot examples to guide the model on expected style and structure
- Increasing --ndocs to 10 and --max_depth to 3–5 to allow more information to influence the generation tree
- Continue fine-tuning Contriever on real AML Q&A pairs

Conclusion

While large language models like GPT-4 show strong retrieval capabilities through prompting, they are not a substitute for structured and scalable RAG systems. Our findings suggest that integrating GPT-based retrieval into RAG architectures – either as a retriever, reranker, or controller – can improve answer quality while maintaining efficiency, traceability, and control. This opens up a path for hybrid retrieval strategies that balance cost, relevance, and fluency in enterprise QA applications.

References

Asai, A., Wu, Z., Wang, Y., Sil, A., & Hajishirzi, H. (2023). Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. arXiv preprint arXiv:2310.11511. <https://doi.org/10.48550/arXiv.2310.11511>